# Ben Bubnick

## Master Career History

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## Career Overview

Interdisciplinary data engineer, analyst, software developer, and educator with a foundation in physics and fine arts and more than a decade of experience building reliable data systems in healthcare, insurance, and color science. Leads ambiguous, high-stakes work from source discovery and requirements through modeling, automation, validation, migration, documentation, and production handoff, spanning cloud data platforms, more than 100 billion clinical records, multi-organization EHR integrations, revenue-cycle models, and applied machine learning. Known for turning complex technical and scientific problems into reproducible systems, clear explanations, and measurable delivery improvements.

Career experience also includes college physics and astronomy teaching, published astrophysics and printmaking research, independent product development and consulting, visual arts, community service, organizational leadership, and hands-on farm and operations work. Across these settings, combines analytical rigor, creative problem solving, practical craftsmanship, continuous learning, and a commitment to making knowledge, tools, and opportunities more accessible to others.

## Skills and Domains

**Programming and Query Languages:** *Current/Core:* SQL, Python, Ruby, Bash/Zsh; *Working Knowledge:* JavaScript, C#, Swift, Java, R; *Historical:* VB.NET, C++, FORTRAN, Perl, IDL, MQL, XPath, XQuery, JSONPath, Jaql

**Application Development:** *Working Knowledge:* APIs and web services, JSON and schema validation, document extraction and OCR-to-LLM parsing, native macOS development, SwiftUI, application permissions and accessibility configuration, scripted packaging, Spring Boot/Jersey; *Historical:* ASP.NET, MVC, Web APIs, React, Node.js, HTML5, CSS, LINQ, commercial-insurance rating applications

**Databases and Query Engines:** *Current/Core:* Snowflake, PostgreSQL, Amazon Athena, Amazon Redshift; *Working Knowledge:* Azure Synapse Analytics, Microsoft SQL Server and T-SQL, MySQL, MongoDB/PyMongo, SQL query tuning and indexing, partitioning and clustering; *Historical:* Oracle and PL/SQL, Hive, Impala, Greenplum

**Data Architecture and Engineering:** *Current/Core:* dbt, SQLMesh, Jinja-templated SQL, Snowpark, ETL/ELT, data integration and transformation, data warehouses and lakes, greenfield data-platform architecture, proof-of-concept development and productionization, source-system discovery and feasibility analysis, source-to-target mapping, dimensional modeling, business-grain definition, natural/composite/surrogate keys, identity crosswalks, metadata-driven transformations, data contracts and lineage; *Working Knowledge:* Databricks Lakehouse, Lakeflow Connect, Lakeflow Spark Declarative Pipelines, Spark, Kafka, Data Vault modeling, Parquet, OMOP Common Data Model; *Historical:* Hadoop/HDFS, Pig

**Data Quality, Governance, and Migration:** *Current/Core:* data-quality strategy and automated testing, acceptance criteria, null/duplicate/referential-integrity and business-rule checks, source-to-target and aggregate reconciliation, schema-drift detection, parity checks, regression testing, UAT, migration-acceptance testing, deidentification validation, root-cause analysis and defect remediation, idempotent and deterministic reruns, audit logging; *Working Knowledge:* Databricks data-governance fundamentals, Unity Catalog interoperability, catalog management and data organization, data privacy, governance at scale, role-based access control (RBAC) design, terminology mapping, data-sharing and consent constraints

**Data Operations and Reliability:** *Current/Core:* job scheduling, workflow orchestration, pipeline and software dependency management, observability, monitoring and alerting, event-driven workflow notifications, incident classification and response, production troubleshooting, runbooks, release and promotion gates, configuration and version management; *Working Knowledge:* Lakeflow Jobs, Databricks workload deployment and performance optimization

**Cloud, DevOps, and Test Tools:** *Current/Core:* AWS (S3, Glue, Lambda, SNS, event notifications, CloudWatch, AWS CLI), Microsoft Azure (Data Factory, Functions, Blob Storage), Git/GitHub, GitHub Actions, Docker, CI/CD, automated unit and integration testing, test fixtures and golden files, RSpec, RuboCop, Liquibase, Make, CLI environment setup, Python virtual environments; *Working Knowledge:* Databricks declarative automation bundles and data-engineering DevOps patterns, hermetic tests, Azure DevOps, Jenkins, Helm/Kubernetes, Gradle, Postman, regular expressions, Tesseract OCR, RubyLLM; *Historical:* cron

**Healthcare Data, Interoperability, and Risk Adjustment:** *Current/Core:* clinical, payer, eligibility, claims, revenue-cycle, and financial data; EHR/EMR integration; HL7 v2/v3, FHIR, EDI/837, electronic visit verification (EVV); patient/provider identity, population health, care gaps, HEDIS and other quality measures and registries, prior authorization, health-information exchange, ADT notifications, PHI handling and HIPAA-compliant deidentification workflows; Medicare risk adjustment and risk capture, RAF/HCC workflows, ICD-10-to-HCC mapping, diagnosis-submission lineage, chart-chase data preparation, MMR/MOR/MAO-004 and payer/CMS response reconciliation; *Working Knowledge:* social determinants of health, provider directories, patient consent, HITRUST readiness support

**Analytics, Machine Learning, and BI:** *Current/Core:* data analysis, statistics, machine learning, predictive modeling, pandas, Jupyter, Power BI, data visualization, dashboard and report design, BI visual standards, operational and executive reporting; *Working Knowledge:* model monitoring, Tableau, R/Shiny, time-series forecasting and ARIMA diagnostics, optimization, NLP and topic modeling, XGBoost, random forests, neural networks, classification and regression evaluation (confusion matrices, ROC/AUC, calibration); *Historical:* SAS/GRAPH

**Scientific Research and Computing:** *Current/Core:* reproducible analysis and research-data workflows; *Working Knowledge:* experimental design, measurement uncertainty, instrument calibration, numerical analysis, linear algebra, mathematical modeling and simulation, finite-difference methods, interpolation, numerical integration and differentiation, error and stability analysis; *Historical:* physics and astrophysics research, telescope observing plans and logs, laboratory equipment maintenance, infrared spectroscopy and photometry, astrometry and source cross-matching, stellar classification and isochrone comparison, optical measurement and spectrophotometry, color matching and formulation, radiative-transfer modeling, statistical risk analysis, Fourier analysis, Monte Carlo methods, IRAF, SpeXtool, DAOPHOT, PASCO DataStudio, Mathematica, MATLAB, LabVIEW, Maple, PDL, JMP

**Teaching and Learning:** *Current/Core:* technical and scientific writing, documentation, stakeholder communication, mentoring, public speaking, training, and translation of complex technical and quantitative concepts for varied audiences; *Working Knowledge:* curriculum and instructional design, recorded/online instruction, rubric design, learning-management systems, accommodations and accessibility practices (presentation, response, timing, and setting), formal evaluation of learning outcomes, research publication; *Historical:* physics and astronomy instruction, laboratory instruction, tutoring and test preparation, peer instruction, Just-in-Time Teaching, Interactive Lecture Demonstrations, student feedback-based instructional improvement, order-of-magnitude estimation and comparative-risk communication

**Consulting, Leadership, and Business:** *Current/Core:* requirements facilitation, client discovery, stakeholder alignment, cross-functional delivery, technical specifications, code review, team mentoring and onboarding, proposal writing, pricing, invoicing, contract negotiation, bookkeeping, accounting, tax compliance, marketing, client acquisition, promotional-material development; *Working Knowledge:* product prototyping, cost and throughput tracking, change management, intellectual-property portfolio support

**Customer Service, Sales, and Administration:** *Working Knowledge:* customer relationship management (CRM), conflict de-escalation, copyediting to a style guide; *Historical:* consultative selling, telephone outreach, point-of-sale transactions, cash reconciliation, returns processing, merchandising, inventory counts, library reference service and collection inventory, Library of Congress classification, editorial writing and community journalism, club administration and event planning

**Community and Civic Service:** *Working Knowledge:* volunteer scheduling, community-health data preparation; *Historical:* public-record data transformation, community meal distribution and guest service, youth mentoring and outreach

**Creative and Visual Arts:** *Historical:* relief, intaglio, screen, and monoprint processes; plate and screen preparation, proofing, editioning and print exchanges; painting, drawing, photography and papermaking; exhibition preparation, commissioned work, studio collaboration, process experimentation and safer-material substitution; Adobe Illustrator, Adobe Photoshop, GIMP, AutoCAD

**Farm, Property, Logistics, and Industrial Operations:** *Current/Core:* orchard establishment, pruning, irrigation, pest monitoring, harvest planning, seasonal tree care, grounds maintenance, supplies and equipment management; *Historical:* estimating and quoting maintenance jobs, surface preparation and painting, concrete repair, ladders and power tools, construction cleanup, warehouse picking/packing and loading, production-line preparation and shop-floor safety, scrap sorting, event setup/teardown, waste and recycling handling

**Food Service and Hospitality:** *Historical:* food preparation, meal-service coordination, coffee and espresso preparation, counter service, dish-room and work-area sanitation, stocking, opening and closing duties, Person in Charge (PIC) certification

**General Tools:** *Current/Core:* Microsoft Office, LaTeX; *Working Knowledge:* Jira, Confluence, Salesforce CRM

## Professional Experience

**HHAeXchange** Oct. 2025 - Jun. 2026
**Senior Data Engineer**
*Technologies: Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase, GitHub Actions, FHIR/Aidbox*
Brought in to unify two distinct data environments following the HHAeXchange-Sandata merger. Built shared Snowflake/dbt models and onboarding patterns, then identified and remediated data-quality defects affecting hundreds of millions of EVV records. Led healthcare source analysis, data-quality investigation and remediation, and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use. Contract via Veersa Technologies.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, establishing reusable patterns for faster client onboarding.
- Diagnosed and remediated a multi-payer EVV dashboard data-loss defect by tracing fact-to-dimension joins across 1.12 billion visit records; quantified 214.5 million records (19.2%) excluded by missing patient, service-code, and payer mappings, isolated 99.2% of in-scope missing-patient visits to the CareSource feed, and corrected the underlying integration defects.
- Executed structural, row-level, transformation, and completeness QA for a 156-field Snowflake/dbt EVV visit model; uncovered and corrected defects associated with a 52.23% source-to-target row-count gap, 73.6% null patient identifiers, and 105.6 million orphaned visit relationships.
- Validated and remediated an EVV patient-address model, resolving six documented defects involving deletion handling, multi-address coverage, usage-tag logic, and production data types; verified 1.31 million unique keys, zero missing parent references, and exact 158,084-row parity for the deduplicated parent-matched population.
- Conducted feasibility analysis of source-system structures and implementation constraints to shape integration scope, modeling strategy, and delivery approach.
- Translated ambiguous business requirements into documented mappings, acceptance criteria, onboarding standards, and 12 data-quality investigation runbooks.
- Defined surrogate-key, natural-key, and crosswalk strategies supporting stable identity resolution across related source systems.
- Structured target models using FHIR-like concepts where appropriate to improve consistency and extensibility for downstream integration and analytics.
- Implemented reusable SQL/Jinja transformation logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains.
- Created or materially redesigned 30 automated SQL/dbt data-quality tests covering nulls, duplicates, referential integrity, business rules, lifecycle logic, and source-to-target reconciliation.
- Collaborated with client and internal stakeholders to resolve source ambiguity, refine requirements, and translate business needs into production-ready data assets.
- Resolved data-quality and mapping issues through targeted test-case analysis and review of failing source and target records.
- Established reusable patterns for onboarding additional sources and extending integration workflows.

**GrowData Analytics** Sep. 2024 - Oct. 2025
**Principal ML Engineer**
*Technologies: Python, Ruby, RubyLLM, RSpec, RuboCop, Tesseract OCR, LLM APIs, Docker, AWS, GitHub Actions*
Founded GrowData Analytics to explore how AI led analytics could help small local farms make better operational decisions. The work exposed the more fundamental obstacle that the source data was trapped in inconsistent invoices. I built a preproduction document-to-data product with OCR, validation, audit trails, operational analytics, and a lightweight AWS deployment.

- Built and validated a preproduction document-to-data MVP that converted semi-structured invoices into auditable transaction records, achieving 95% field-level accuracy across a stratified 500-invoice test suite.
- Instrumented the MVP with automated tests, audit logging, retry controls, and confidence routing, achieving 88% straight-through processing and under 1% system failure.
- Added schema checks, cross-field validation, reconciliation, idempotent reruns, fixtures, golden-file tests, linting, and release automation.
- Productized Tesseract OCR and rule-based post-processing as a small Python/Ruby service with containerized deployments and versioned configuration.
- Built a lightweight analytics layer tracking usage, error categories, latency, and throughput to guide feature tuning and cost control.
- Led marketing and client acquisition, including creating promotional materials for the business.
- Managed infrastructure as code for a minimal AWS footprint using S3, Lambda or scheduled jobs, and CloudWatch alerts through Bash/Zsh and Makefile workflows.
- Documented runbooks and failure modes covering timeouts, low-confidence OCR, and retries.
- Managed the business's bookkeeping, accounting, and tax obligations.

**Adonis** Mar. 2025 - Jul. 2025

**Senior Data Engineer**

*Technologies: AWS S3, Snowflake, SQLMesh, SQL, Python, Docker, GitHub Actions*

Joined a short-turnaround engagement to onboard Hopebridge Autism Therapy Center onto a revenue-cycle data platform. Transformed poorly documented source data into 54 tested Snowflake models, resolved fundamental key usage problems, and delivered a reusable onboarding and validation framework. Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding, analytics, and reporting. Contract via Health Data Movers.

- Diagnosed undocumented source-key versioning and implemented composite business grains with surrogate keys across 10 charge and payment models, eliminating duplicate-key failures.
- Built, documented, validated, and maintained 54 SQLMesh models that delivered standardized revenue-cycle datasets for a provider whose network served 4,598 families across 113 locations and processed 86,388 annual authorizations.
- Implemented automated row-count, uniqueness, and not-null audits across 54 models, achieving a 100% pass rate across 117+ configured audit applications at final delivery.
- Built a reusable SQLMesh/Snowflake ingestion and transformation framework using templated models, data contracts, CI checks, and promotion gates.
- Standardized transformation logic across clients and translated business requirements into technical specifications to improve delivery predictability.
- Authored implementation playbooks, technical handoff documentation, and a QA runbook for a repeatable healthcare-data validation workflow.
- Added schema-drift alerts, parity checks, reconciliation logs, deterministic rerun procedures, and incident-handling guidance for UNIX-like environments.
- Packaged a lightweight operations forecasting and alerting pattern as a tested Python and Docker service with GitHub Actions deployment automation.

**Arkos Health** Jun. 2023 - Jan. 2025

**Senior Data Analyst**

*Technologies: AWS S3, Athena, Redshift, Glue, SNS, dbt, PySpark, Docker, Jupyter, Power BI, Tableau*

Joined as a senior data analyst supporting Medicare risk capture, then grew into the data engineer and platform builder the team needed. Built a greenfield AWS data-lake proof of concept using Glue, Athena, event notifications, and SNS. After validating the approach, designed and built the final architecture spanning the data lake and Redshift, using dbt as the transformation and modeling layer.

- Architected and built a greenfield AWS data-lake foundation integrating 18 source systems through approximately 200 scheduled data assets and 200 dbt models, supporting six analysts across six state markets; CI/CD automation and targeted model refactoring accelerated deployments and improved query performance.
- Configured event notifications for the greenfield data-lake CI/CD workflows, routing workflow status and failure events through Amazon SNS for centralized tracking.
- Established SNS topics and subscriptions for CI/CD workflow notifications, giving the analytics team timely visibility into deployment outcomes and supporting follow-up on failed runs.
- Built and maintained the risk-adjustment SQL/dbt model inventory; supported three RAF notebooks, their companion workbook, and the Tableau Risk Dashboard while preserving an auditable record of workflow logic, inputs, outputs, and provenance.
- Built the SQL/dbt ingestion and normalization layer used by version-aware AWS Glue and Athena pipelines for evolving MAO, HCC, MOR, and related payer files, preserving source lineage and standardizing downstream datasets in partitioned Parquet across the lakehouse.
- Built SQL/dbt models for multi-market Medicare RAF data preparation in Arizona, North Dakota, and Illinois, combining enrollment and MMR demographic factors with medical claims, signed EHR diagnoses, and payer encounter and MAO diagnoses; generated member and diagnosis inputs supporting CMS-HCC scoring and pre-/post-wellness-visit RAF analysis.
- Built SQL/dbt models implementing diagnosis-submission lineage and acceptance logic covering add/delete activity, repeated submissions, encounter disposition, source and date-of-service preservation, and selection of the latest accepted diagnosis state.
- Built SQL/dbt reconciliation models that traced diagnoses from signed EHR encounters through Waystar claims, supplemental files, and MAO-004-style payer and CMS responses, enabling the risk team to isolate where diagnoses dropped from the submission path.
- Built SQL/dbt models for diagnosis-level HHS-HCC capture analytics that mapped ICD-10 diagnoses from signed face-to-face visits, reconciled them with payer gap reports, classified existing, new, and net-new capture, identified the first capture source, and produced monthly capture and HCC-per-visit measures.
- Built SQL/dbt models and datasets for supplemental diagnosis, claims, and member-enrollment workflows serving Allwell, NextBlue, Humana, Allymar, and Reveleer, joining claims to the appropriate member, MBI, enrollment period, contract, demographic segment, and member-month and formatting outputs to downstream specifications.
- Built a dbt-based Reveleer chart-chase data product across three Medicare contracts, covering eligible-member and encounter selection, market-specific rules, provider and chart-location resolution, stable chase identifiers,

and generation of separate chase and diagnosis files.
- Built multi-market SQL/dbt datasets combining member, diagnosis, MOR, payer-gap, and ICD-to-HCC mapping data to support analysis of the CMS-HCC V24-to-V28 transition.
- Integrated existing Jupyter-based analytics and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring, executive, and BI reporting.
- Instrumented scheduled pipelines with monitoring, alerting, and more than 10 failure classifications; authored three AWS/dbt incident runbooks supporting rapid detection and recovery.
- Expanded cross-departmental collaboration to bring operational context into analytics decisions.
- Initiated recurring stakeholder-feedback sessions to refine deliverables against business needs.
- Mentored engineers on code review, testing, and pipeline design and authored internal delivery standards.
- Led goal-setting sessions to establish shared objectives and a consistent tracking cadence.
- Implemented Jira and GitHub workflows to centralize task tracking and improve coordination across the analytics team.
- Mentored junior team members in code review, testing, pipeline design, and delivery practices.

**Merative** Jul. 2022 - Mar. 2023

**Senior Data Scientist**

*Technologies: Microsoft Azure, Azure Data Factory, Blob Storage, Azure Functions, SQL Database, Synapse Analytics, Python, SQL, Azure DevOps*

Served as the quality and acceptance layer for a large scale healthcare platform migration from HDFS to Azure. Built automated reconciliation, regression, deidentification, and governance controls that brought the tested data to sign-off with no unresolved discrepancies.
- Led HIPAA-compliant deidentification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.
- Engineered a seven-layer Python and SQL migration-acceptance suite with zero unresolved discrepancies at sign-off across tested record counts, partitions, schemas, nulls, duplicates, field values, aggregates, and business rules.
- Enforced data-governance and security protocols supporting audit-risk reduction and HITRUST readiness.
- Collaborated with multidisciplinary engineering teams to refine architectural designs and preserve workflow continuity and data fidelity.
- Validated end-to-end Azure Data Factory pipelines, Blob Storage configurations, and downstream Synapse integrations during the transition from HDFS to Azure.
- Designed and implemented automated ETL testing and regression suites that detected schema drift and transformation errors early in the migration lifecycle.
- Documented test plans, platform-migration procedures, and onboarding guidance for future quality-assurance engineers.

**IBM** Jun. 2015 - Jul. 2022

Progressed through five increasingly senior roles over seven years, evolving from hands on healthcare data integration into technical leadership across platform engineering, data architecture, governance, analytics, and cloud migration validation. Repeatedly converted bespoke client problems into reusable tooling, governed platforms, validation frameworks, and team practices.

*Technologies: Spark, Hadoop/HDFS, Hive, Impala, Kafka, Python, SQL, T-SQL, MySQL, PostgreSQL, PL/SQL, MQL, XPath, XQuery, Apache Pig, Jaql, JSONPath, Docker, Jenkins, Tableau, HL7*

**Senior Data Scientist** Nov. 2021 - Jul. 2022

Led healthcare data integration, migration validation, and analytics delivery across clinical, claims, operational, and financial sources, including an HDFS-to-Azure transition spanning more than 150 billion deidentified records.
- Served as technical lead across electronic health records, insurance claims, scheduling and financial data, social determinants of health, and other relevant sources.
- Developed migration-validation analysis spanning more than 150 billion deidentified health records, reconciling source and target schemas, partitions, values, aggregates, and business rules during the transition from legacy HDFS infrastructure to Azure.
- Supported data and integration work for Pennsylvania's statewide P3N health-information exchange, encompassing master patient identity, provider directories, clinical-document discovery, patient consent, and real-time ADT encounter notifications.
- Engineered a Spark, Hive, and Impala analytics framework powering 50+ prior-authorization lag-time and approval-bottleneck reports with daily operational and periodic regulatory and financial snapshots.
- Worked with senior leaders and colleagues to identify opportunities, define objectives, and formulate analytics strategies that produced measurable impact.
- Served as a lead member of Provider Analytics, performing source discovery, data profiling, and exploratory analysis while integrating disparate EHR and centralized data sources into analytics-ready data marts.
- Supported the controlled decommissioning of legacy Explorys capabilities by analyzing downstream dependencies and validating continuity across normalized clinical, claims, laboratory, billing, and operational datasets.

- Supported Cleveland Clinic Foundation's accelerated SuperMart migration from MySQL to Greenplum by assessing dashboard dependencies and defining the changes required to preserve reporting continuity.
- Led delivery of a tailored analytics and reporting project using client administrative data.
- Provided production and configuration support for Phytel population-health applications, troubleshooting data and operational issues affecting provider configuration, care-gap identification, patient outreach, and quality-reporting workflows.
- Created and delivered Phytel configuration training covering provider activation, system changes, support procedures, and the clinical-data workflows underlying proactive population-health management.
- Developed and delivered peer-led Azure instruction through the REACH initiative, translating cloud architecture, storage, data integration, security, and analytics concepts into practical guidance for technical colleagues.
- Performed postmigration validation for an enterprise Jira transition, reconciling projects, issues, workflows, custom fields, permissions, histories, and relationships and documenting discrepancies for remediation before cutover.

**Senior Data Architect** Dec. 2020 - Nov. 2021

Directed data-onboarding architecture and governance for large-scale clinical platforms, combining metadata-driven integration, deidentification, lineage, validation, and applied machine learning.

- Served as a subject-matter expert to project data architects and delivery, analytics, and integration teams; directed configuration of the data-onboarding platform and supporting infrastructure.
- Replaced static mappings with metadata-driven instruction sets across approximately 200 source integrations, standardizing EDI batch schemas while enabling Python/Jupyter lineage tracing and end-to-end validation gates.
- Architected Spark and Hadoop pipelines integrating more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of measures and registries.
- Led a HIPAA patient-deidentification project drawing from more than 100 data sources for content analytics and measures.
- Led a cross-functional team as the clinical data analyst for the 2020/2021 measure-updates project.
- Served as a senior member of Data Onboarding & Governance, integrating healthcare data, enriching metadata, and configuring data models, analytics, reports, and visualizations.
- Served in an advisory capacity for development of a new data platform.
- Developed a topic-modeling project using nearly six million semi-structured provider notes.
- Developed a surgical-scheduling optimizer and CNN/RNN case-card prediction proof of concept that was being evaluated as a potential new offering.
- Created an exploratory machine-learning COVID-19 prototype to surface hospitalization-risk signals for retrospective analyst review.
- Closed project-documentation gaps by organizing material from more than 100 past projects.

**Data Architect** Nov. 2017 - Dec. 2020

Led healthcare-data onboarding and validation across complex multi-EHR integrations, strengthening data-sharing controls, patient-matching quality, and repeatable client delivery.

- Led a multi-organization healthcare integration spanning 70+ EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules and coordinating delivery with engineering teams.
- Designed SQL validation frameworks and client-review gates across approximately 100 integrations, reducing Jira-tracked validation rework by 25% and mapping rework by 45%.
- Built the data-onboarding platform and associated infrastructure, remediated UAT data issues, and reported on data health during onboarding.
- Earned AI certification and peer-reviewed patient-matching algorithm tuning across the project portfolio.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.34%.
- Received the Lab Services Award for coordinating internal project teams to deliver client needs on time.
- Curated 25 pages of AI-workflow documentation and mentored new hires in the AI-tuning process.
- Spearheaded a new-hire mentoring program that standardized technical ramp-up and prepared colleagues to contribute to billable client delivery.
- Served on tiger teams supporting the HDFS-to-Azure data-platform upgrade and participated on the data-governance team.
- Played a central role in the REACH cross-team training initiative, teaching SQL fundamentals to nontechnical colleagues and ETL, architecture, data-science, and predictive-modeling practices to technical colleagues.
- Supported employee onboarding from 2016 to 2018, helping new colleagues learn team practices, systems, and technical workflows.

**Software Developer, Data Platform** Sep. 2016 - Nov. 2017

Developed healthcare data-platform tooling and training programs that automated HL7 ingestion, reduced vendor costs, and accelerated knowledge transfer.

- Engineered an HL7 v2/v3 parsing pipeline in Pig for approximately 30 internal users, replacing a Mirth-based workflow and eliminating $30,000 in annual vendor costs.
- Received a GDP bonus for improving revenue per SaaS client by at least 1.17%.
- Received the Eminence and Excellence Award in 2017 for an analytics project.

- Rewrote and reorganized project documentation to streamline knowledge transfer.
- Redeveloped the employee education program to make new-hire training more efficient and repeatable.
- Participated in internal coding jams, hackathons, and cross-team delivery initiatives.

**Data Scientist** Nov. 2015 - Sep. 2016

Directed client data-integration delivery and built reusable extraction and QA practices that reduced implementation effort and standardized onboarding across dozens of healthcare projects.

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Served as principal integration specialist on more than 60 projects for 20 clients across the broader 2015–2018 data-ingestion period.
- Built a Python, Ruby, and SQL extraction tool that eliminated unnecessary story points and reduced average engineering hours per client integration.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.
- Eliminated an average of three weeks from employee onboarding.

**Explorys, an IBM Company** Jun. 2015 - Nov. 2015

**Data Scientist**

*Technologies: Hadoop/HDFS, Ruby, Pig, React, Node.js, JavaScript, SQL, Oracle, Jenkins*

Joined Explorys during a period of rapid client expansion and its transition into IBM Watson Health. Helped scale healthcare data onboarding across more than 10 client warehouses by leading source discovery and integration setup, diagnosing Hadoop and Oracle ingestion failures, and introducing scheduling, recovery, and blackout controls for production data feeds. Contract via Ranstad USA.

- Identified and extracted data from heterogeneous client systems; led technical requirements gathering, discovery, documentation, and implementation; and coordinated acquisition, installation, and configuration of VPN hardware supporting client integrations.
- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.
- Developed a cron-based control for defining blackout periods around client data pulls.

**AmTrust Financial Services** Mar. 2014 - Jun. 2015

**Software Engineer**

*Technologies: VB.NET, ASP.NET, C#, Web APIs, CSS, JavaScript, HTML5, SQL, PL/SQL, LINQ, R*

Brought in to develop agent facing commercial fire insurance quoting and policy generation software. Grew into the principal developer for a new product offering, translating ISO rating rules into more than 65 web forms, APIs, and SQL services while strengthening testing, performance, architecture, and coordination between business analysts and developers.

- Implemented a new line of business in VB.NET and Microsoft SQL Server, served as principal developer for a new product offering, and contributed to two major software-architecture initiatives in 2015.
- Developed more than 65 ISO rating-based web forms and analyzed page timings in R to identify optimization targets and improve application performance.
- Supported the technical integration of acquired businesses into the commercial policy platform.
- Served as the working liaison between business analysts and developers, translating requirements, reporting progress, and helping prioritize delivery needs.
- Evaluated the advantages, risks, and implementation costs of requested features and recommended maintainable solutions grounded in object-oriented design principles.
- Designed, developed, tested, and maintained full-stack commercial-insurance applications spanning VB.NET and ASP.NET interfaces, multi-tier components, Web APIs, and SQL data services.
- Created a unit- and integration-test suite for multi-tier MVC components, improving regression coverage and establishing stronger team testing practices.
- Coordinated developer schedules and troubleshot production problems across multiple application environments.
- Refactored and maintained the existing VB.NET insurance-rating codebase to improve readability, performance, and long-term maintainability.

**PPG Industries** Apr. 2012 - Aug. 2013

**Color Scientist**

*Technologies: VB.NET, C#, SQL, C++, FORTRAN, JMP, spectrophotometry, numerical methods*

Joined PPG as a color scientist and software specialist supporting the modernization of legacy automotive color-matching technology. Applied spectroscopy, radiative-transfer physics, statistical modeling, and numerical methods to improve matching and measurement consistency, then became a global liaison helping translate that work into software, standardized workflows, training, and operational adoption.

- Served as technical voice for color data-management software, supporting its global development, launch, training, adoption, and application.
- Served as the global liaison among Color, Technical, Operations, and IT teams, translating color-science requirements into consistent systems, workflows, and communications.

- Championed a more science-centered organizational approach to color by introducing analytical methods, shared standards, and evidence-based decision-making.
- Developed statistical models from established theory, experimental data, and novel approaches to improve color measurement, matching, and process decisions.
- Evaluated color processes, instrumentation, and data flows to identify technical and operational risks and develop practical mitigation strategies.
- Managed a portfolio of color-related intellectual property and supported the translation of research into operational tools and practices.
- Led cross-functional color-science and marketing initiatives from concept through on-time, in-full delivery for the Automotive Refinish business.
- Improved color-matching classification accuracy using numerical integration methods.
- Built cross-instrument correlation software in VB.NET and JMP to improve measurement consistency and support statistical risk analysis.
- Cowrote color workflows and best practices for internal operations and external customers, supported the change-management efforts required for adoption.
- Informed marketing strategy and communications by connecting customer needs with technical capabilities and broad operational insight.
- Wrote a Visual C++ and VB.NET DLL wrapper for a loaned spectroscopic instrument and developed formulation methods for an automated color-matching facility.
- Debugged a VB.NET and FORTRAN optimization routine for a numerical radiative-transfer solution.

## Selected Independent Projects

**ql2-sixt-canada-analysis** 2026
Python and Jupyter analytical project evaluating whether an hourly Sixt Canada rental-car rate feed is trustworthy enough to support customer decisions. Builds reusable ingestion and governance contracts, fail-closed data-quality and pricing-readiness gates, authority-backed location canonicalization, and temporal reconciliation before analyzing matched airport-versus-downtown premiums, hourly price-change events, and visible-assortment stability. Protects proprietary source data and derived results, validates the implementation with synthetic-data pytest suites, and delivers six reproducible notebooks culminating in a final report, a monitoring blueprint, and a six-slide presentation with value-level traceability.

**iphone-cam-lite** 2025 - 2026
Native Swift/SwiftUI macOS menu bar application that selects an iPhone Continuity Camera and microphone in Zoom or Microsoft Teams, then reads back the application state to verify that the requested devices were actually selected. Features modular application adapters, permission-aware onboarding, system diagnostics, configurable device and accessibility labels, persistent settings, actionable error reporting, hermetic Swift tests, continuous integration, and scripted application packaging.

**farm-invoice-parser** 2025
Ruby command line prototype that converts handwritten or printed farm invoices into structured transaction records. The pipeline extracts text from invoice images with Tesseract OCR, sends the result to configurable language-model providers through RubyLLM, parses the response as JSON, and validates it against a versioned invoice schema. It also provides an OCR-only diagnostic mode and includes RSpec, RuboCop, and GitHub Actions scaffolding.

**cfd_snow_fence** 2025
Python applied modeling experiment for evaluating potential snow fence locations using measured wind data. The project maps wind vectors onto a two-dimensional spatial grid, estimates pressure fields through a finite difference Navier–Stokes/Poisson calculation, and aggregates the resulting simulations. It then trains an XGBoost classifier using position, wind speed, and pressure to identify low wind regions as a proxy for potential snow drift accumulation.

**claims_case_study** 2023
End to end healthcare workforce planning case study analyzing Cleveland area clinician, facility, and claims data for 2021–2022. The project evaluates source data quality, converts the original workbook into schema preserving Parquet datasets, standardizes the analytical data, and performs exploratory and time series analysis. ARIMA forecasts with error diagnostics project 2023 demand for Internal Medicine, Urology, and Anesthesiology, culminating in a written technical report and stakeholder oriented presentation.

**HL7-ingestion** 2023
Java 21 Spring Boot/Jersey service that accepts HL7 v2 messages as multipart file uploads or plain text and converts them into JSON encoded FHIR R4 collection Bundles using the LinuxForHealth converter. The project packages the service with Docker and a Helm chart for Kubernetes deployment, includes GitHub Actions automation, and exercises transformation and error handling through tests that validate generated Patient, Encounter, Observation, Practitioner, and AllergyIntolerance resources.

**DETL-design-challenge** 2023
Architectural design for a dynamic ETL system that transforms heterogeneous clinical research data into the OMOP Common Data Model. The design combines detailed source to target specifications, human readable transformation rules, generated SQL, and explicit review, testing, and debugging stages so that common mappings can be automated without removing expert oversight from complex semantic decisions. It also documents data quality controls, terminology mapping, de-identification considerations, and implementation options involving Apache NiFi, Azure Data Factory, and Python.

**covid-dataset** 2020 - 2022
Azure Data Factory data engineering project that ingests public ECDC COVID-19 and population data over HTTP and Blob Storage, lands raw data in ADLS Gen2, transforms cases, deaths, and hospital admissions with mapping data flows, and publishes curated snapshots to Azure SQL Database. Implements metadata-driven, scheduled, and event-driven ingestion; country, population, and calendar enrichment; idempotent, failure-safe staged SQL publication with audit records and concurrency controls; managed-identity and least-privilege security; ARM deployment templates; and continuous integration checks for secrets, ADF references, T-SQL, and generated artifacts.

**zams_cluster_candidacy** 2019 - 2020
Python scientific computing pipeline for evaluating stellar cluster candidacy from Washington system photometry. The analysis filters observations by photometric uncertainty and cluster radius, incorporates decontamination membership probabilities, and corrects color–magnitude diagrams for reddening and distance. It then traces candidate cluster sequences with kernel density contours and compares them with zero age main sequence and isochrone models across age and metallicity ranges, producing per cluster and aggregate diagnostic plots.

**quality_measures_engine** 2018 - 2019
Python/PyMongo prototype for defining and evaluating population health quality measures from normalized patient records. It represents measure metadata, parameters, calculated dates, initial populations, denominators, numerators, exceptions, and exclusions as reusable JSON definitions, then translates those definitions into MongoDB queries or map reduce operations. Fixtures and automated tests demonstrate the design using sample NQF cervical cancer screening and pneumonia vaccination measures.

**gamma_rays** 2019
Python and Jupyter machine learning research project for reconstructing events from high energy gamma ray telescope data. It reads simulated FITS event tables, balances gamma ray and hadronic background samples, and compares random forest, multilayer perceptron, and exploratory neural network classification approaches using confusion matrices, ROC/AUC, Q-factor, and calibration metrics. The project also trains regression models for event energy and direction reconstruction and includes reusable prediction code, persisted models, technical reports, and disseminated notebook analyses.

**Star-Chart** 2015
Browser based HTML, CSS, and JavaScript astronomy application that renders the visible night sky for a selected location and date. It accepts browser geolocation or manually entered coordinates, calculates Julian date and local sidereal time, projects catalog stars according to right ascension, declination, and magnitude, and computes positions for the Sun, Moon, and visible planets. Stars, planets, and optional constellation lines are drawn on an interactive, zoomable canvas.

## Professional Recognition

**Special Recognition, Health Data Movers - Client Delivery** Jul. 2025
Recognized for diagnosing undocumented source-key versioning in a revenue-cycle integration and implementing composite business grains with deterministic surrogate keys across 10 charge and payment models, eliminating duplicate-key failures and stabilizing client delivery.

**Arkos Health BI Star** Nov. 2023
Internal Business Intelligence recognition received during a period marked by establishing the team's GitHub workflow, delivering 27 BI reports, creating the Arkos Power BI theme and visual standards, and strengthening PHI deidentification and analytics documentation supporting value-based-care operations.

**Manager Recognition for Transition Leadership, Merative - Systems Migration** Feb. 2022
Recognized by management for transition leadership while assuring the HDFS-to-Azure platform migration: built a seven-layer Python and SQL acceptance suite, validated Azure Data Factory, Blob Storage, and Synapse workflows, and led HIPAA-compliant deidentification testing across data representing approximately 300 million patient lives, reaching sign-off with zero unresolved discrepancies.

**IBM Champion Learner** 2016 - 2022
Progressed from Bronze in 2016 to Silver in 2017, then earned Gold annually from 2018 through 2022 through IBM Your Learning continuing education, completing 1,477 learning hours and earning 43 credentials. Gold recognition required at least 200 annual learning hours and three new badges or certifications.

**Lab Services Award, IBM - Cross-Team Delivery Coordination** Jul. 2018
Recognized for leading a multi-organization healthcare integration spanning more than 70 EHR sources, implementing provider-scoped patient filtering to enforce data-sharing rules, and coordinating engineering and delivery teams to meet client commitments.

**Eminence and Excellence Award, IBM - Data-Curation Tool Development** Jul. 2017
Recognized for engineering an Apache Pig pipeline that parsed HL7 v2 and v3 data for approximately 30 internal users, replacing a Mirth-based workflow and eliminating $30,000 in annual vendor costs.

**Color Lab Automation Award, PPG Industries - Expanded Reporting** Sep. 2012
Recognized as part of the Color R&D team supporting PPG's new computer-controlled automotive-refinish color-matching laboratory in Cleveland; expanded automated data collection and analysis reporting used to evaluate pigments and increase formulation throughput, precision, and operational efficiency.

**Mary J. Hanna Research Fellowship, University of Cincinnati** Aug. 2010
Department of Physics fellowship providing financial support for completion of M.S. research using near-infrared photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk star clusters.

**Academic Scholarships and Honors** 2000 - 2010
Received University Graduate Scholarships supporting physics study from 2008 to 2010; earned University of Cincinnati Dean's List recognition from 2005 to 2007; received Ohio University's faculty-endorsed Provost's Undergraduate Research Fund grant for the 2003 low-toxic photo-etch project and earned Dean's List recognition from 2002 to 2004; received the Ohio Proficiency Scholarship in 2000 after passing all five sections of the state's twelfth-grade proficiency examination.

## Teaching and Academic Research

Across these institutions, used interactive, peer-led, and just-in-time teaching Built an interdisciplinary foundation in scientific research, quantitative analysis, and education before entering the technology industry. Conducted published astrophysics and experimental printmaking research, developed calibrated and reproducible data products, and taught physics and astronomy across laboratory, technical, general education, and online settings. Established a teaching practice centered on evidence, accessibility, interactive learning, and continuous improvement—skills later applied to technical leadership, documentation, mentoring, and stakeholder communication.

**Lorain County Community College** Aug. 2016 - May 2019
**Adjunct Professor** - Taught the laboratory component of algebra-based General Physics I (then PHYC 151) and the three-credit ASTY 152 Solar System Astronomy lecture course.

- Exceeded institutional student-evaluation benchmarks by an average of 16% in 2017 and 17% in 2018.
- Reduced reported student confusion by 28% by reorganizing online instructional materials and reducing the need for clarification during office hours.
- Developed and taught online astronomy instruction covering astronomical observation, planetary-system formation, terrestrial and Jovian planets, small bodies, and space exploration.
- Used Just-in-Time Teaching strategies to adapt instruction for underperforming students based on pre-class responses.
- Developed and published more than 70 hours of publicly available recorded lectures and laboratory content to support flexible, on-demand learning.
- Guided students through laboratory work in mechanics, energy, momentum, rotation, and thermal physics, emphasizing measurement uncertainty, quantitative analysis, and scientific reporting.

**Cincinnati State Technical and Community College** Feb. 2011 - Jun. 2011
**Adjunct Professor** - Developed and delivered the integrated Technical Physics I–III lecture and laboratory sequence for technical-program students with varied mathematics preparation.

- Wrote laboratory instructions and coordinated team-based measurement, data-analysis, and technical-report work.
- Delivered two weekly lecture hours and three laboratory hours per course across applied mechanics, thermodynamics, electricity and circuits, waves, and optics.

**Miami University** Aug. 2010 - Jun. 2011
**Adjunct Professor** - Developed and taught three-credit general-education courses PHY 101 Physics and Society and PHY 111 Astronomy and Space Physics.

- Designed a concept-first applied-physics curriculum for non-science majors, translating advanced technical subjects into accessible, decision-ready frameworks without relying on calculus.
- Structured lessons around consequential real-world questions, teaching students to distinguish scientific constraints from policy and value judgments and to communicate evidence-based conclusions to nontechnical audiences.
- Taught students to use order-of-magnitude estimates, energy density, power, conservation laws, and comparative risk to evaluate claims involving transportation fuels, electricity generation, climate, and emerging technologies.
- Connected nuclear physics and radioactivity with power generation, medical imaging, radiation risk, and nuclear security; linked gravity, waves, and the electromagnetic spectrum with satellites, GPS, radar, communications, and remote sensing.
- Taught astronomy topics spanning planetary systems, stellar evolution, galaxies, cosmology, and the scientific search for extraterrestrial life.

**University of Cincinnati** Jan. 2006 - Sep. 2010
Held concurrent research and teaching appointments in physics, astronomy, and engineering computation.

**Research Assistant, Department of Physics** Sep. 2006 - Sep. 2010

Conducted near-infrared astronomical research using observations from the 0.8–5.4 micrometer SpeX spectrograph at the NASA Infrared Telescope Facility.

- Combined spectral classifications with 2MASS photometry to estimate cluster extinction ($A_V \sim 21$–$23$), distance ($1.0$–$1.75$ kpc), age of no more than a few million years, and mass of approximately $10^3$ solar masses.
- Helped classify the cluster's massive stellar population, identifying an O5 V dwarf, several early-B stars, and a massive young stellar object exhibiting CO band-head emission from dense circumstellar material.
- Published the resulting 130-source astrometric and photometric catalog through the international VizieR astronomical data service.
- Co-developed an astrometrically calibrated catalog of 130 sources by registering SpeX acquisition images to 2MASS; compiled available $JHK$ photometry and used DAOPHOT to separate blended sources in the crowded field.
- Cross-matched infrared and radio-source positions and identified candidate near-infrared counterparts for three deeply embedded radio sources associated with early star formation.
- Reduced and analyzed $0.8$–$2.4$ micrometer, $R \sim 2000$ SpeX spectra and measured diagnostic H I, He I, He II, C IV, N III, and CO spectral features across nine candidate cluster members.
- Estimated stellar radial velocities by measuring pixel-space shifts of CO absorption features against normalized standard-star spectra.
- Acquired follow-up L-band spectra of Galactic OB stars with SpeX during three consecutive nights at NASA's IRTF.
- Performed astronomical spectroscopy, photometry, classification, calibration, signal analysis, and noise reduction using IRAF, IDL, SpeXtool, Mathematica, and Perl.

**Teaching Assistant** Jan. 2006 - Jun. 2010

Led approximately 60 laboratory meetings for more than 100 engineering, science, and preprofessional students in the algebra/trigonometry-based College Physics sequence (PHYS 111–113) and calculus-based General Physics sequence (PHYS 211–213).

- Designed interactive laboratory instruction using Peer Instruction and Interactive Lecture Demonstrations to identify and address student misconceptions in real time.
- Taught experimental design, measurement uncertainty, data acquisition, graphical analysis, and technical reporting using PASCO DataStudio and Microsoft Excel.
- Supported group recitations in freshman engineering physics through demonstrations, short lectures, and individual assistance.
- Served as teaching assistant and grader for the corresponding College Physics lecture sequence (PHYS 101–103) and General Physics lecture sequence (PHYS 201–203).
- Supported Astronomy and the Natural Universe lectures (PHYS 121–122) and astronomy laboratories (PHYS 125–127) through grading and instructional assistance.

**Engineering Teaching Assistant** Oct. 2009 - Dec. 2009

Prepared and delivered Mathematica instruction for Numerical Methods for Engineering Design, an upper-division aerospace-engineering course.

- Connected computational methods with aerospace analysis and design problems, showing students how to translate mathematical formulations into executable workflows and interpret the resulting output.
- Reinforced numerical accuracy, error, stability, and method selection so students could evaluate the credibility of computed engineering results.
- Developed step-by-step demonstrations applying symbolic and numerical computing to interpolation and approximation, numerical integration, nonlinear equation solving, systems of equations, and ordinary differential equations.

**Ohio University** Oct. 2003 - May 2004

**Student Researcher, Fine Arts** - Earned a competitive Provost's Undergraduate Research Fund award to adapt a wintergreen-oil toner-transfer process from lithography to intaglio photo-etch printing.

- Replaced xylene- and acetone-based photo-transfer steps with methyl salicylate, eliminated the need for specialized exposure equipment, and demonstrated a process using less than $20 in materials.
- Developed and tested a complete copper- and zinc-plate workflow spanning surface preparation, asphaltum grounding, high-contrast toner transfer, burnishing, clearing, aquatint, and staged etching.
- Characterized critical process variables and failure modes, including ground thickness and cure time, solvent saturation, image resolution, transfer pressure, evaporation time, and temperature control during aquatint fusion.

## Publications and Research Outputs

**Applied Research and Technical Writing**

Bubnick, Ben. (2025). "Reproducible Postgres load of Tuva seed datasets." Internal applied research series, GrowData Analytics/HSEA. Document tracking no. 20250401-A. Internally published by HSEA, Aug. 2025. Author's version, Apr. 1, 2025.

Bubnick, Ben. (2020). "Surgical Scheduling Optimization and Procedure Duration Prediction." Internal project report, IBM. Document tracking no. RC25712.

Bubnick, Ben. (2017). “Process and Requirements for CCD Data Integration.” Internal white paper, IBM. Document tracking no. RC25043.
Bubnick, Ben. (2013). “Characterization of Surface Coatings Using RT Models for Color Matching.” Unfiled patent application authored for PPG Industries and retained by the company as a trade secret. Ref. 1284 - February 2013.
Bubnick, Ben. (2010). “The B Stars of Westerlund 1.” Unpublished internal stellar-spectroscopy analysis, University of Cincinnati.

**Journal Articles and Data Catalogs**
Bubnick, Ben. (2015). *AI paint formula prediction from spectroscopic measurements* (Version 1.2) [Interactive web application]. shinyapps.io.
Bubnick, Ben. (2010). *Massive stellar clusters in the disk of the Milky Way Galaxy* [Master’s thesis, University of Cincinnati]. OhioLINK Electronic Theses and Dissertations Center. OhioLINK record
Hanson, M. M., & Bubnick, Ben. (2008). *JHK photometry in cluster [BDS2003] 107* [Data set]. VizieR Online Data Catalog (Catalog J/PASP/120/150). `https://doi.org/10.26093/cds/vizier.61200150`
Hanson, M. M., & Bubnick, Ben. (2008). Near-infrared spectroscopy of candidate members of the Galactic cluster [BDS2003] 107. *Publications of the Astronomical Society of the Pacific, 120*(864), 150–159. https://doi.org/10.1086/527403
Bubnick, Ben. “A Novel Method for Low-Toxic Photo-Etch Printing.” *MAPC Journal*, vol. 11, no. 2, Fall/Winter 2003, pp. 12-13.

## Speaking Engagements

**Surgical Scheduling Optimization and Procedure Duration Prediction - IBM AI Hackathon - Dec. 2020**
Presented a prototype that combined historical operating-room utilization and the current surgical wait list to optimize scheduling and improve total procedure-time estimates beyond traditional fixed-ratio methods; outlined CNN/RNN case-card prediction as a path toward a potential new offering.
**IBM Watson Care Insights - IBM REACH Initiative - 2019**
Delivered conceptual and hands-on training with synthetic patient data, explaining how longitudinal records, real-time HL7 feeds, and FHIR configuration support clinical-data integration and the delivery of insights within physicians’ EHR workflows.
**Machine Learning with Real-World Health Data - Content and Data Analytics Learning Exchange - 2018**
Introduced an implementation-oriented approach to applying machine-learning algorithms to real-world health data, connecting clinical-data preparation, model development, evaluation, and interpretation.
**Natural-Language Processing for Unstructured Data Lakes - Innovation Research Interchange Fall Networks Conference - Sep. 2018**
Presented a healthcare case study using natural-language processing and topic modeling to turn nearly six million semi-structured provider notes in a data lake into analyzable themes and features.
**Hadoop Ingestion and Data-Lake Concepts - Cross-Team Training Series - 2016**
Designed and led a seven-part series connecting hands-on Hadoop/HDFS ingestion workflows with distributed storage, transformation, validation, troubleshooting, and the use of data lakes for heterogeneous clinical data.
**Massive Stellar Clusters in the Disk of the Milky Way Galaxy - M.S. Astrophysics Thesis Defense, University of Cincinnati - Jun. 2010**
Defended research using near-infrared broad-band photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk clusters and estimate their distance, age, and approximate mass.
**Astronomical Research Using 2MASS Data - 2MASS Research Poster Session - 2008**
Presented results from combining 2MASS $JHK$ photometry with IRTF/SpeX observations to characterize candidate members of the young Galactic cluster [BDS2003] 107, including its stellar population, extinction, distance, age, and approximate mass.
**A Novel Method for Low-Toxic Photo-Etch Printing - Ohio University Research Poster Session - 2004**
Presented Provost’s Undergraduate Research Fund-supported work adapting a wintergreen-oil toner-transfer process from lithography to intaglio photo-etch printing, including a low-cost copper- and zinc-plate workflow that replaced xylene- and acetone-based transfer steps and eliminated specialized exposure equipment.

## Creative and Visual Arts

**Media:** Relief printmaking (woodcut and linocut), intaglio (etching, drypoint, photo-print, and aquatint), screenprint, monoprint, monotype, and papermaking; oil and acrylic painting; graphite, pencil, charcoal, ink, marker, pastel, and mixed-media drawing; photography.

**Selected Exhibitions**
**Drawn and Quartered VII, HEDGE Gallery, 78th Street Studios, Cleveland, Ohio** May 15–Jun. 5, 2015
**Exhibiting Artist** - Participated in a group exhibition of drawings. **Works:** *Molly I* (graphite and pastel on paper, 2013; retained); *Molly II* (graphite and pastel on paper, 2013; gifted); *Untitled* (pencil on paper, 2014; retained); *Untitled* (pencil on paper, 2014; retained); *Untitled* (marker on paper, 2015; retained); *Amanda in 6 colors* (pencil on paper, 2014; retained); *Amanda in 3 colors* (pencil on paper, 2014; retained); *Keri* (pencil and pastel on paper, 2015; gifted); *Emily* (pencil and pastel on paper, 2012; retained); *Student at the CMA* (pencil on paper, 2014; retained); *Girl from Treehouse* (pencil and pastel on paper, 2013; retained); *Girl from the Meadow* (pencil and pastel on paper, 2013; retained); *Chatty Nurse on the Bus* (pencil and pastel on paper, 2013; retained).

**13th Annual Holiday Store, Heights Arts, Cleveland Heights, Ohio** Nov. 6–Dec. 31, 2014
**Exhibiting Artist** - Presented limited-edition photography and oil paintings in the annual holiday exhibition and sale. **Works:** *Berea Falls* (photograph, 2014; edition of 20, 2 artist's proofs; 7 sold, remainder retained); *Road near Boston* (photograph, 2014; edition of 15, 1 artist's proof; 4 sold, remainder retained); *Lakewood Park* (photograph, 2013; edition of 12, 1 artist's proof; 3 sold, remainder retained); *Brandywine Falls* (photograph, 2013; edition of 25, 2 artist's proofs; 9 sold, 1 gifted, remainder retained); *Rocky River Reservation* (photograph, 2014; edition of 10 sets, 1 artist's-proof set; 2 sets sold, remainder retained); *CMA Gardens* (photograph, 2013; edition of 10, 1 artist's proof; 2 sold, remainder retained); *Cardinal I* (photograph, 2013; edition of 12, 1 artist's proof; 3 sold, remainder retained); *Cardinal II* (photograph, 2014; edition of 20, 2 artist's proofs; 7 sold, remainder retained); *Lake Erie I* (oil on canvas, 2014; retained); *Lake Erie II* (oil on canvas, 2014; sold); *Lake Erie III* (oil on canvas, 2014; retained).

**Holiday Off The Wall Members Show and Sale, Zygote Press, Cleveland, Ohio** Dec. 6–7, 2014
**Exhibiting Artist** - Presented relief prints and monoprints in the annual members' exhibition. **Works:** *Elise* (woodcut, 2004; edition of 20, 2 artist's proofs; 2 sold, remainder retained); *Struggle* (woodcut, 2004; edition of 20, 2 artist's proofs; retained); *Girl I* (monoprint, 2013; sold); *Girl II* (monoprint, 2013; retained); *Girl III* (monoprint, 2013; sold); *Girl V* (monoprint, 2013; sold).

**Waterloo Arts Fest, Collinwood, Cleveland, Ohio** Jun. 29, 2013
**Exhibiting Artist and Printmaker** - Presented limited-edition prints and pulled an open-edition festival print on site for visitors. **Works:** *Lake Erie* (linocut, 2013; edition of 25, 2 artist's proofs; 4 sold, remainder retained); *Lake Erie, Low Water* (woodcut, 2013; edition of 12, 1 artist's proof; 3 sold, remainder retained); *Breakwall* (drypoint, 2012; edition of 10; 2 sold, remainder retained); *Transit* (etching and aquatint, 2012; edition of 15; 2 sold); *Waterloo (festival print)* (linocut pulled on site, 2013; open edition of roughly 40; given away at the booth).

**Tremont ArtWalk, Tremont, Cleveland, Ohio** 2013
**Exhibiting Artist** - Presented relief, intaglio, screenprint, monotype, and charcoal work in a group exhibition. **Works:** *Buckeye Carbon* (woodcut, 2013, first state; edition of 10; 1 gifted, remainder retained); *Viaduct* (etching, 2013; edition of 12, 1 artist's proof; retained); *Night Shift, East 30th* (charcoal on paper, 2013; retained); *Analemma* (screenprint, 2013; edition of 20; retained); *Steam Plant* (monotype, 2013; retained).

**Holiday Off-the-Wall Exhibition, Zygote Press, Cleveland, Ohio** Dec. 1–2, 2012
**Exhibiting Artist** - Presented relief prints and monoprints in the annual members' exhibition. **Works:** *Elise* (woodcut, 2004; edition of 20, 2 artist's proofs; retained); *Struggle* (woodcut, 2004; edition of 20, 2 artist's proofs; retained); *Girl I* (monoprint, 2013; retained); *Girl II* (monoprint, 2013; retained); *Girl III* (monoprint, 2013; retained); *Girl IV* (monoprint, 2013; sold); *Girl V* (monoprint, 2013; retained).

**Juried Exhibition, Zygote Press, Cleveland, Ohio** Jun. 1–Jul. 7, 2012
**Exhibiting Artist** - Presented photo-print etchings in a juried exhibition. **Works:** *Transit of Venus, 5 June 2012* (photo-print etching, 2012; edition of 15, 2 artist's proofs; retained); *Ephemeris (plate I)* (photo-print etching, 2012; edition of 10, 1 artist's proof; retained).

**Neighborhood Artists, Kennedy Heights Arts Center, Cincinnati, Ohio** 2010
**Exhibiting Artist** - Participated in a group exhibition of paintings by neighborhood artists. **Works:** *Montgomery Road, Rain* (oil on panel, 2010; gifted); *Back Lot, Kennedy Heights* (oil on canvas, 2010; gifted); *Basin Fog* (oil on canvas, 2009; gifted); *Ludlow Viaduct* (acrylic on panel, 2009; gifted); *Night Kitchen* (oil on panel, 2010; gifted).

**Final Friday Open Studios, Pendleton Art Center, Cincinnati, Ohio** 2007–2008
**Studio Artist** - Presented paintings from a shared studio during recurring open-studio events. **Works:** *Mt. Adams from Columbia Parkway* (oil on canvas, 2007; sold); *Over-the-Rhine Window* (oil on panel, 2007; sold); *Basin Fog (first version)* (oil on canvas, 2007; gifted); *Eden Park Overlook* (oil on panel, 2008; sold); *Pendleton Studio, Afternoon* (oil on panel, 2008; gifted); *Viaduct at Dusk* (acrylic on panel, 2008; gifted).

**B.F.A. Thesis Exhibition, Jennings House Gallery, Ohio University, Athens, Ohio** May 2004
**Exhibiting Artist** - Presented senior printmaking work spanning relief, intaglio, and screenprint. **Works:** *Self Portrait I* (woodcut, 2003; edition of 10, 2 artist's proofs; retained); *Self Portrait II* (woodcut, 2004; edition of 10, 2 artist's proofs; retained); *Self Portrait III* (woodcut, 2004; edition of 10, 2 artist's proofs; retained); *Brick Kiln* (etching and aquatint, 2004; edition of 12, 1 artist's proof; retained); *Coal Seam* (aquatint and drypoint, 2004; edition of 8, 1 artist's proof; retained); *Court Street, 2 a.m.* (screenprint, 2004; edition of 20, 2 artist's proofs; retained).

**Grand Opening, Lacuna Gallery, Athens, Ohio** Apr. 2004
**Exhibiting Artist** - Displayed an original screenprint in the gallery's opening exhibition. **Work:** *Court Street, 2 a.m.* (screenprint, 2004; edition of 20; retained).

**Undergraduate Juried Exhibition, Ohio University Art Galleries, Athens, Ohio** 2004
**Exhibiting Artist** - Presented three prints in the university's juried exhibition. **Works:** *Brick Kiln, Nelsonville* (etching and aquatint, 2004; edition of 12, 1 artist's proof; retained); *Ridge and Hollow II* (woodcut, 2004; edition of 10, 2 artist's proofs; retained); *Court Street, 2 a.m.* (screenprint, 2004; edition of 20, 2 artist's proofs; retained).

**Student Print Exchange, Ohio University Art Galleries, Athens, Ohio** Feb. 2004
**Organizer and Exhibiting Printmaker** - Organized a student print exchange, editioned a work to the size of the exchange, and exhibited it alongside the received prints. **Work:** *Company Houses (exchange impression)* (etching, 2004; edition of 22 for the exchange, 2 artist's proofs; 20 distributed to participants, artist's proof retained).

**Print Exchanges, Conferences, and Commissions**

**Mid America Print Council Conference Print City: Detroit 2014, Wayne State University, Detroit, Michigan** Sep. 24–27, 2014
Contributed a woodcut to a conference exchange portfolio. **Work:** *Buckeye Carbon (exchange impression)* (woodcut, 2014; edition of 30, 2 artist's proofs; 28 distributed through the portfolio, artist's proof retained).
**Southern Graphics Council Conference Making Histories: Revolution and Representation, Boston, Massachusetts** Apr. 2–5, 2003
Contributed a linocut to a student exchange portfolio. **Work:** *Night Loading Dock (exchange impression)* (linocut, 2003; edition of 25, 2 artist's proofs; 24 distributed through the student portfolio, artist's proof retained).
**Ohio University School of Art Commission** 2003
**Commissioned Artist** - Created an original woodcut commissioned by the School of Art; a commissioned impression was presented to Raymond Tymas-Jones, Dean of the College of Fine Arts. **Work:** *Tree on Paper* (woodcut, 2003; edition of 10, 1 artist's proof; commissioned impression gifted).

**Studio Access, Workshops, and Visiting-Artist Projects**
**Zygote Press, Cleveland, Ohio** 2011–2014
Completed independent studio work in etching, relief, and screenprint.
**Morgan Art of Papermaking Conservatory, Cleveland, Ohio** Summer 2012
Completed a papermaking workshop.
**Manifest Drawing Center, Cincinnati, Ohio** 2009–2010
Participated in open figure-drawing sessions.
**Art Academy of Cincinnati Community Education** Summer 2005
Completed oil-painting instruction in Cincinnati, Ohio.
**Debra Fisher Visiting-Artist Print Project, Ohio University** Summer 2003
**Studio Assistant** - Supported plate and chine colle preparation, proofing, and editioning.
**Dan Murphy Visiting-Artist Print Project, Ohio University** Summer 2003
**Studio Assistant** - Supported screen preparation, proofing, and editioning.

## Volunteer and Community Experience

**HSEA** Feb. 2025 - May 2025
**ETL Brigade Volunteer** - Built a reusable PostgreSQL data model and reproducible loading process for Tuva seed datasets, enabling community-health organizations to prototype analytics using claims, visits, and eligibility data. Added schema setup, normalization, and validation components to make the data consistently loadable and testable.
**CSforAll** Oct. 2018 - May 2019
**Instructor** - Supported computer science instruction for Cleveland Metropolitan School District high school students as part of a district wide effort to make rigorous computer science coursework available in every high school. Helped broaden access for students from underrepresented and low-income populations, with particular attention to learners without reliable computer or Internet access at home.
**HIT in the CLE** Mar. 2017 - Mar. 2019
**Coach** - Coached five-student high school teams through a data science competition, helping participants analyze a large dataset, develop evidence-based conclusions, and communicate their findings. Introduced students to health-information-technology careers while building data analysis and presentation skills and supporting a more diverse Northeast Ohio Health IT talent pipeline.
**Code for America** Jan. 2015 - Jun. 2015
**ETL Guild Volunteer** - Applied ETL and data science methods to initiatives that made difficult to use public records more accessible and machine-readable for research, planning, and civic use. Supported broader efforts to improve government transparency and help residents engage with public information.
**St. Herman House - FOCUS Cleveland, Cleveland, Ohio** Sep. 2013 - Mar. 2014
**Daily Meal Program Volunteer** - Supported a year-round meal program providing three meals a day to neighbors in need, helping deliver welcoming, dignity centered service.
- Distributed meal tickets, plated food, handed meals to guests, and served water and desserts in the dining room.

**Heights Observer, Cleveland Heights, Ohio** Feb. 2012 - May 2014
**Volunteer Editor** - Edited resident-submitted reporting for a nonprofit, citizen-produced newspaper serving Cleveland Heights and University Heights, improving accuracy and clarity while preserving each contributor's voice. Also contributed local arts and culture coverage, supporting civic participation, public discourse, and independent community journalism.
**The News Record, University of Cincinnati** Feb. 2008 - Sep. 2010
**Editorial Contributor** - Wrote and edited campus content for the university's independent, student-run newspaper. Helped inform and educate the university community, amplify student perspectives, and maintain accurate, timely coverage of campus life.
**TCP World Academy** Sep. 2008 - May 2009
**Tutor** - Tutored students in support of TCP World Academy's mission to develop academically engaged, independent learners. Reinforced classroom learning through technology and participatory micro-society activities that connected students' education with meaningful participation in a global society.
**Project Astro at Ethel M Taylor Academy** Nov. 2008
**Volunteer Astronomer** - Worked with an educator at Ethel M Taylor Academy to provide hands-on, inquiry based astronomy and physical science activities. Helped students investigate scientific questions and engage with astronomy as practicing scientists through the national Project ASTRO educator-astronomer partnership model.

**Sigma Theta Epsilon, Beta Gamma Chapter, Cincinnati, Ohio** 2006 - 2008
**Chapter Volunteer** - Participated in volunteer service through the University of Cincinnati chapter of an interdenominational Christian fraternity.

**Reach Out on Campus, Ohio University, Athens, Ohio** Sep. 2000 - Jun. 2004
**Volunteer** - Supported community meal services through the Christian campus ministry during undergraduate study.
- Routinely served meals and cleaned up at weekly Tuesday community meals at Athens Church of Christ.
- Participated in on-campus meal services open to the public.

## Leadership, Memberships, and Activities

**Zygote Press, Cleveland, Ohio** 2011 - 2014
Member and independent studio artist working in etching, relief, and screenprint. Zygote is a nonprofit community printmaking studio, gallery, archive, and education center that expands artists' access to affordable professional workspace, instruction, exhibitions, residencies, and collaborative development.

**Detroit Color Council** 2012 - 2013
Member of a nonprofit industry forum promoting communication and education among engineers, designers, scientists, and other professionals involved in automotive color and appearance. The council advances shared technical knowledge and practices intended to improve the quality of colored and decorative materials.

**American Institute of Aeronautics and Astronautics Cincinnati Student Chapter** 2009 - 2011
Member of a student chapter connecting participants with the aerospace profession and AIAA's emphasis on innovation, technical excellence, and professional collaboration. Participated in Students for the Exploration and Development of Space in 2009, supporting its mission to advance student involvement in space exploration through educational and engineering projects.
- Vice President, 2010.

**University of Cincinnati Astronomy Club** 2008 - 2010
Founding member in 2008 who helped establish a student community intended to cultivate interest in astronomy through observation, education, and shared exploration of the night sky.
- Secretary, 2009–2010.
- Treasurer, 2008–2009.

**Additional Academic Activities** 2006 - 2010
Participated in the Large Synoptic Survey Telescope seminar group, UC SPARKS, and University of Cincinnati physics departmental seminars, which provided forums for research discussion and scientific exchange. Also participated in the University of Cincinnati Mountaineering Club, which promotes safe wilderness skills, outdoor appreciation, stewardship, and community through instruction and group outings.

**Society of Physics Students Cincinnati Chapter** 2005 - 2008
Helped lead a chapter of the national professional association for physics students, which builds supportive scientific communities and promotes education, research, professional development, outreach, and public understanding of physics.
- Treasurer, 2006–2007.
- Secretary, 2005–2006.

**Sigma Theta Epsilon Beta Gamma Chapter** 2006 - 2007
Member of an interdenominational Christian service fraternity organized around faith, fellowship, service, and meeting unmet human needs.
- Led planning for the Beta Gamma chapter's 10th-anniversary banquet.
- Chaplain, 2006–2007 academic year.

**Mid America Print Council** 2003 - 2004
Student member of an educational and community-based organization that supports traditional and contemporary printmaking through conferences, exhibitions, publications, professional exchange, and research. Its stated values emphasize inclusion, diversity, integrity, and creative expression.

**Southern Graphics Council (now SGC International)** 2002 - 2004
Student member of an educational nonprofit that advances printmaking and graphic arts through professional exchange, exhibitions, publications, conferences, and recognition of artists. The organization promotes research, public appreciation, creative dialogue, and participation across diverse communities.

**Athens Wargamers** 2001 - 2004
Member of an Ohio University student community centered on tabletop wargaming, strategic problem solving, and social play.
- Treasurer, 2002–2003 academic year.

**Reach Out on Campus, Ohio University** Sep. 2000 - Jun. 2004
Member of a Christian campus ministry at Ohio University focused on fellowship, spiritual growth, campus outreach, and community service.
- Student Leader, 2003–2004 academic year.

## Academic Services

**Huntington Learning Center** Feb. 2012 - Aug. 2013
**Math/Science Exam Prep Teacher** - Provided individualized tutoring and test preparation, specializing in ACT and SAT exams while also supporting students in math, science, and other academic areas.

- Adapted instruction to individual learning needs across math, science, and additional coursework beyond college-entrance exam preparation.
- Taught exam content and test-taking strategies through targeted practice, helping students strengthen subject knowledge, pacing, and confidence for the ACT and SAT.

**University of Cincinnati Geology Math Physics Library** Mar. 2006 - Sep. 2008
**Student Librarian** - Supported research and collection operations for a specialized university library.

- Assisted students, faculty, and other patrons with research and reference needs and helped locate relevant materials.
- Maintained Library of Congress-classified collections, conducted inventory, and investigated missing items.

**University of Cincinnati Learning Communities Program** Jun. 2006 - Aug. 2008
**Peer Leader** - Facilitated a learning community for approximately 25 first-year students co-enrolled in multiple courses, supporting their academic, social, and professional transition to university life.

- Provided individual and small-group mentoring, helping students form study networks, navigate university systems, and connect with faculty, advisers, and academic-support services.
- Planned community-building activities and coordinated with faculty, advisers, program staff, and campus organizations to strengthen belonging, peer accountability, and academic engagement.
- Prepared and facilitated twice-weekly meetings on course assignments, study strategies, time management, academic expectations, campus resources, and first-year transition needs.

**Private Tutoring** Jun. 2006 - Aug. 2007
**Private Tutor** - Provided sustained one-on-one instruction to a student completing the calculus-based introductory physics sequence, PHYS 201–203.

- Diagnosed misconceptions through guided questioning and targeted practice, reinforcing difficult concepts with step-by-step explanations and alternative solution strategies.
- Taught a repeatable problem-solving framework built around diagrams, governing physical principles, dimensional analysis, estimation, and interpretation of calculated results.
- Adapted short lectures, demonstrations, and representative worked problems to the student's individual learning needs and progress across the three-course sequence.

**University of Cincinnati** Summer 2006
**Supplemental Instruction Leader** - Supported an intensive summer program in which academically advanced incoming students completed the full calculus-based introductory physics sequence, PHYS 201–203, before beginning the regular academic year.

- Helped students connect concepts across the compressed three-course sequence through diagrams, dimensional analysis, estimation, and physical interpretation of calculated results.
- Planned and led collaborative review sessions for groups of up to 12 students, integrating physics content with transferable learning strategies, addressing misconceptions, and modeling successful student behaviors.
- Delivered focused problem-solving lectures between formal class meetings, working representative problems step by step and demonstrating efficient solution strategies.

## Property Operations and Manual Labor

**Doodlebug Farms** Sep. 2024 - Present
**Family Farm Operator** - Establishes and operates a pre-production orchard with 72 American plum and Chickasaw plum trees.

- Manages seasonal tree care, grounds maintenance, supplies and equipment, and physically demanding outdoor work across the developing orchard.
- Plans staged expansion to approximately 216 trees, three times the current planting, while preparing the orchard to transition from establishment into harvest and sales.

**Independent Contractor, Cleveland, Ohio** Jun. 2003 - Apr. 2015
**Residential Painter and Maintenance Contractor** - Completed home-improvement and property-maintenance projects.

- Prepared interior surfaces, applied paint, protected surrounding finishes, and maintained orderly work areas through project completion.

**Cleveland Convention Center, Cleveland, Ohio** Aug. 2013 - Jan. 2014
**Event Setup Crew** - Set up and broke down trade-show spaces at downtown convention venues.

- Assembled booths, laid carpet, installed pipe and drape, and cleared materials after events.

**Cleveland-Area Distribution Warehouse** Aug. 2013 - Jan. 2014
**Warehouse Fulfillment Worker** - Picked and packed orders for shipment at a distribution facility south of Cleveland.

**Downtown Cleveland Construction and Redevelopment Sites** Jan. 2012 - Mar. 2012
**General Laborer** - Performed construction cleanup and material handling on downtown projects.

- Cleared debris and moved materials at job sites around Public Square and on the Flats East Bank redevelopment.

**Independent Contractor, Cleveland, Ohio** Summer 2003
**Concrete Repair Laborer** - Repaired architectural concrete flooring and residential driveways for home-improvement and maintenance projects.

- Prepared damaged areas, placed and finished repair material, and completed job-site cleanup to restore worn concrete surfaces.

**Cleveland-Area Metal-Stamping and Parts Supplier** Summer 2003
**Production Floor Laborer** - Packed finished metal parts and handled scrap on a small industrial production floor.
- Fed scrap bins and kept finished parts moving through packing and material-handling steps.

**Brook Park Warehouse, Ohio** Winter 2003
**Truck Loader and Unloader** - Loaded and unloaded trucks at a warehouse near Cleveland Hopkins International Airport.

**Cleveland-Area Scrapyard and Harbor Boneyard** Winter 2003
**Scrap Sorting and Yard Laborer** - Sorted metal at a scrapyard near the Cuyahoga River and performed manual labor at a harbor boneyard.

**Cleveland-Area Residential Construction Sites** Summer 2002
**Construction Cleanup Laborer** - Cleaned houses under construction in suburban developments including Strongsville and North Royalton.
- Hauled scrap drywall, swept finished houses, and loaded dumpsters with construction debris.

**Cleveland-Area Moving, Property Clearance, and Demolition Assignments** Summer 2002
**General Laborer** - Assisted local movers, cleared foreclosed houses, and worked for several weeks on a crew removing an old plant on Cleveland's East Side.

**Gund Arena and Jacobs Field, Cleveland, Ohio** Summer 2002
**Event Setup and Teardown Crew** - Prepared venues for concerts and games and cleaned up after large crowds.
- Moved folding chairs and staging, dismantled event setups, and completed post-event cleanup.

**Crown Metal Finishing** Summer 2002
**Production Laborer** - Supported an industrial chrome-plating operation through hands-on material handling and line preparation.
- Loaded metal components onto production racks for machine-applied chrome plating, arranging parts for consistent processing and efficient line flow.
- Sustained production pace during repetitive manual work and followed shop-floor handling and safety procedures around plating equipment.

**Gravity Games** Summer 2002
**Event Operations Crew** - Supported venue sanitation and waste-handling operations during a large public sporting event.
- Collected and disposed of waste and recycling throughout the venue and removed litter from high-traffic areas during and after the event.

**FedEx** Winter 2002
**Package Handler and Loader** - Performed time-sensitive loading and material-handling work in a high-volume logistics environment.
- Completed physically demanding assignments involving repetitive lifting, rapid material movement, and coordination with other handlers.

## Sales and Customer Service

**Target** Winter 2010
**Sales Associate** - Processed customer purchases and cash and credit transactions in a high-volume retail environment.
- Assisted customers, stocked merchandise, supported inventory activities, and helped resolve transaction concerns.

**FranklinCovey** Jun. 2005 - Apr. 2006
**Productivity Consultant** - Advised customers on planning products, workshops, and productivity needs using detailed product knowledge.
- Matched products and services to individual customer goals and supported cash and credit transactions.

**Dillard's** Aug. 2004 - Jun. 2005
**Sales Associate** - Helped customers evaluate merchandise and complete purchases.
- Supported merchandising, stocking, opening and closing activities, and customer-service issue resolution.

**DialAmerica Marketing** Winter 2002
**Marketing Representative** - Supported telephone-based customer outreach and sales operations.
- Conducted customer outreach and sales conversations by telephone.

## Food Service and Hospitality

**Lincoln Park Pub** Aug. 2013 - Mar. 2014
**Cook** - Supported restaurant kitchen operations in a fast-paced service environment.
- Prepared food and coordinated kitchen work to support accurate, timely meal service.

**Aladdin's Eatery** Summer 2002
**Dishwasher** - Supported dish-room sanitation and restaurant kitchen operations.
- Maintained clean dishes, utensils, equipment, and work areas throughout restaurant service.

**Barnacle Bill's Lobster House** Summer 2001
**Cook** - Supported kitchen production in a fast-paced restaurant environment.
- Prepared food and helped maintain timely meal production during restaurant service.

**Cravings Cafe** Winter 2001
**Barista** - Provided beverage preparation and counter service in a cafe environment.
- Prepared coffee and espresso beverages, served customers, replenished supplies, and maintained the service

area.

**Pier W** Summer 2000

**Cook** - Supported food preparation and restaurant kitchen operations.

- Assisted with food preparation, kitchen cleaning, and meal-service activities.

**Take 5 Coffee Company** Winter 2000

**Barista** - Supported day-to-day coffee-shop and counter-service operations.

- Prepared coffee and espresso beverages, stocked supplies, and completed opening and closing activities.

## Credentials and Continuing Education

**Databricks Academy Credentials** Sep. 2026

Earned 13 Databricks credentials spanning data engineering and data governance.

- **Data engineering:** Automated Deployment with Declarative Automation Bundles (Credential ID 194935119); Databricks Performance Optimization (Credential ID 194934527); Databricks Data Privacy (Credential ID 194933863); Advanced Techniques with Spark Declarative Pipeline (Credential ID 194933166); Build Data Pipelines with Lakeflow Spark Declarative Pipelines (Credential ID 194932264); Deploy Workloads with Lakeflow Jobs (Credential ID 194931356); Data Ingestion with Lakeflow Connect (Credential ID 194928030); and DevOps Essentials for Data Engineering (Credential ID 194930163).
- **Data governance:** Data Interoperability with Unity Catalog (Credential ID 194981221); Data Governance at Scale (Credential ID 194980190); Catalog Management and Data Organization (Credential ID 194979341); Get Started with Data Governance on Databricks (Credential ID 194978447); and Academy Accreditation - Data Governance Fundamentals (Credential ID 194977782).

**Salesforce Trailhead Coursework Completed at Arkos Health** 2023 - 2025

Reached Trailblazer Mountaineer rank with 18,725 points and 40 badges.

- **Tableau and data literacy:** Tableau Desktop, Tableau Cloud, Tableau Prep, visualization, data structures and field types, filtering and sorting, aggregation and granularity, data connections, collaboration, and presentation.
- **Analytics, AI, and security:** CRM Analytics, correlation and regression, data-analytics fundamentals, natural-language processing, generative and responsible AI, Agentforce, and data security.
- **Salesforce administration and CRM:** Salesforce CRM, user management and engagement, Lightning Experience, formulas and validations, data modeling and management, and reports and dashboards.

**Fundamentals of Visualization with Tableau - University of California, Davis** Jun. 2023

Credential ID M3VVMS4BTXWU.

**IBM Data Science Professional Certificate - IBM** 2022

**IBM Digital Credentials and Coursework** Nov. 2017 - Feb. 2022

Completed 43 distinct IBM credentials spanning data science, machine learning, data engineering, cloud, product management, design thinking, security, blockchain, and quantum computing.

- **Data engineering and big data:** Hadoop Foundations and Programming; Hadoop Data Access; Spark; Apache Kafka data pipelines; Big Data Foundations, Levels 1–2; Data Science Tools.
- **Data science and machine learning:** IBM AI Associate Data Scientist; Data Science Foundations, Levels 1–2; Data Science Methodologies; Python, R, Scala, TensorFlow, GPU-accelerated deep learning, data visualization, and machine-learning coursework.
- **Cloud and security:** IBM Cloud Private Infrastructure and Architecture; IBM Cloud Private Foundation Technology; Cloud Core; Cloud Service Management and Operations; Think Like a Hacker; Watson and Cloud Foundations.
- **Product, Agile, and design:** Agile Explorer; AgileTA Explorer; Product Manager curriculum covering product definition, delivery, business management, and scaling; Enterprise Design Thinking Practitioner and Co-Creator.
- **Emerging technologies:** Blockchain Essentials and Blockchain Essentials V2; Quantum Conversations; Watson Visual Recognition; Cognitive Practitioner.

**Certificate of Specialization in Data Science - Johns Hopkins University** 2015

**Additional Coursework** 2012 - 2014

Windows Forms; quantitative finance; information risk; computer networks; computational data analysis; AI planning.

## Education

**M.S. Astrophysics, University of Cincinnati** Dec. 2010

Completed the thesis *Massive Stellar Clusters in the Disk of the Milky Way Galaxy*, using near-infrared photometry and spectroscopy to identify and characterize two dust-obscured Galactic disk star clusters.

**B.F.A. Printmaking, Ohio University** Aug. 2004

Studio focus in intaglio, relief, and screenprint. Completed a senior thesis exhibition and faculty-endorsed Provost's Undergraduate Research Fund work adapting a low-toxic toner-transfer process to intaglio photo-etch printing.